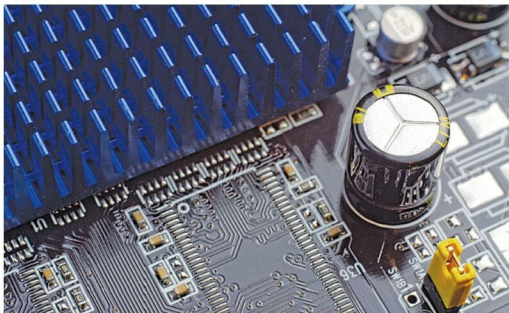


The Golden Rules Of ELECTRONIC CIRCUIT DESIGN



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Knowledge of common pitfalls in designing electronic circuits, and how to avoid these, is important for producing any product



While testing a smartphone, the manufacturer finds a major flaw in the device. Some internal body parts have overheated and melted during the stress-test. A root-cause analysis indicates the designed circuit was not optimised for thermal resilience—a step that should have been planned and executed at the start of the design process. There are many such instances where the solution is simple but gets missed amidst the complexity of a design process. That is why designers jot down the list of best practices to ensure they deliver optimal design with minimal rework and resource. In this article, we take a look into some of these best practices and understand their importance.

Common struggles in circuit design

Designers often face issues of rework and wasted resources. This is mainly due to the use of smaller tracks for circuit layout, where all components are not able to fit. Another reason is the rush to meet timelines.

The design team from PCB Power Market, a PCB design and manufacturing firm from India, says, “In a hurry, designers start miss-

ing out some essential components. So they start using other objects with simple parts and traces and construct replacements for components. It works initially. The layout looks perfect when they view it. But eventually, they give out.”

Communication with the manufacturer is also critical. If the capacity of the manufacturer is not clear, or if the designer has not shared all the components and design schematic with full clarity, the manufactured end product will be flawed. Along with these come the technical challenges.

The golden rules

Pre-design and post-design analyses are equally important to ensure proper circuit outcome. These involve the following considerations.

Choose the right board spacing. Selecting the right spacing can substantially save cost and rework. According to PCB Power team, “Wherever possible, use wider tracks or spacing. If your design permits bigger tracks via drill size, use them. A via-aspect ratio of 6:1 ensures that your board can be